

SMART IRRIGATION USING SOIL-MOISTURE SENSING AND LoRa

Review 2 — System Design

Final Year Project 2025–26

Submitted by

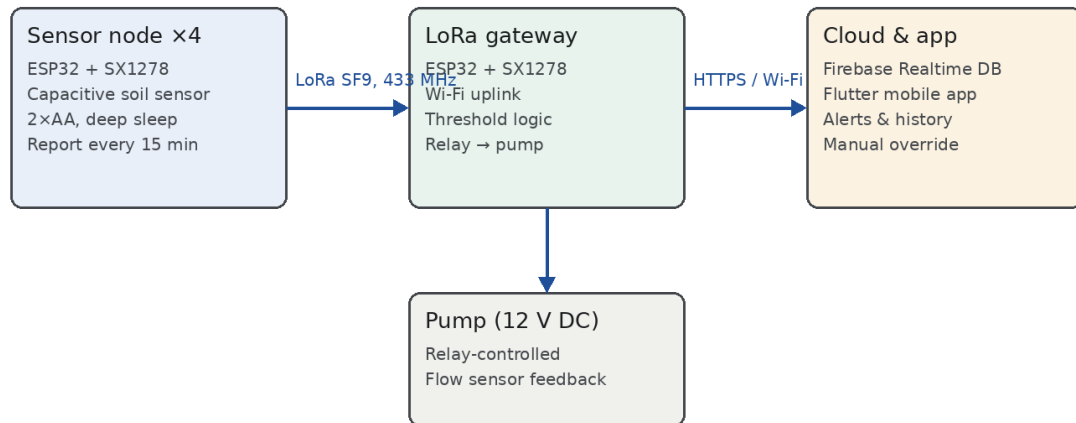
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1. System architecture

Fig. 2.1 System architecture — Smart Irrigation (LoRa)



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Figure 2.1 shows the three tiers of the system: battery-powered sensor nodes, a mains-powered gateway, and the cloud dashboard.

2. Module description

Module	Hardware / software	Responsibility
Sensor node	ESP32, SX1278, capacitive probe	Sample moisture, transmit, sleep
Gateway	ESP32, SX1278, relay	Receive, decide, switch pump
App	Flutter, Firebase	Live readings, history, alerts